

## Highlights

### **The first-passage time density for the diffusion model with variable drift and variable starting point**

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- Closed-form density for the diffusion model with variable drift and starting point
- No closed form exists in the large-time series; the small-time series has one
- Analytic gradients; the seven-parameter likelihood needs one 1-D quadrature
- Verified symbolically, to 48 digits, against three packages, and in Lean 4
- Cheaper than every current package at the settings it fits with

# The first-passage time density for the diffusion model with variable drift and variable starting point

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## Abstract

The Ratcliff diffusion model is fitted with across-trial variability in drift rate, starting point, and non-decision time. The drift integral of the first-passage time density has been available in closed form; the starting-point integral has been evaluated numerically in every implementation, and Tuerlinckx (2004) concluded that no closed form exists. We show that this holds only for the large-time series representation. In the small-time representation the starting-point integral is a Gaussian moment, and we give the density with normally distributed drift and uniformly distributed starting point in closed form, together with a large-time counterpart in terms of the Faddeeva function. All partial derivatives are closed form, and the seven-parameter likelihood needs a single one-dimensional quadrature. The result is verified symbolically, to 48 digits, against three reference implementations and process simulation, and is machine-checked in Lean 4.

*Keywords:* Diffusion model, Response time modeling, First-passage time, Starting point variability

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## 1. Background

The diffusion model for response times (Ratcliff, 1978) describes a two-choice decision as a Wiener process with drift  $v$  and unit diffusion coefficient<sup>1</sup> evolving between two absorbing barriers at 0 and  $a > 0$  from a starting point

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<sup>1</sup>For diffusion coefficient  $\sigma^2 \neq 1$  all results below hold after the substitution  $(v, \eta^2, a) \rightarrow (v/\sigma, \eta^2/\sigma^2, a/\sigma)$  (Blurton et al., 2017).

5  $aw$ ,  $0 < w < 1$ . The first-passage times at the two barriers are the predicted  
6 response times for the two alternatives. To account for the fast and slow  
7 errors observed in most response time experiments, the model is applied  
8 with across-trial variability in three of its quantities (Ratcliff, 1978; Ratcliff  
9 and Rouder, 1998; Ratcliff and Tuerlinckx, 2002): the drift rate is normally  
10 distributed,  $v \sim N(\nu, \eta^2)$ ; the starting point is uniformly distributed,  $w \sim$   
11  $U(w_1, w_2)$ , with Ratcliff's  $s_z = a(w_2 - w_1)$ ; and the non-decision time is  
12 uniformly distributed over a window of width  $s_t$ . The predicted density of a  
13 response time is then a triple integral of the constant-parameter first-passage  
14 time density over these three distributions.

15 Of the three integrals, only the drift integral has been available in closed  
16 form. The first-passage time density with normally distributed drift is given  
17 by Horrocks and Thompson (2004) and, for the lower barrier, reads (Blurton  
18 et al., 2017, Eq. 1)

$$g(t \mid \nu, \eta^2, a, w) = \frac{1}{\sqrt{2\pi t^3 S}} \exp\left[\frac{-\nu^2 t - 2\nu aw + \eta^2(aw)^2}{2S}\right] \\ \times \sum_{j=0}^{\infty} (-1)^j r_j \exp\left(-\frac{r_j^2}{2t}\right), \quad (1)$$

19 where  $S := 1 + \eta^2 t$ ,  $r_j = ja + aw$  for even  $j$  and  $r_j = ja + a(1 - w)$  for odd  
20  $j$ . Blurton et al. (2017) derived the corresponding cumulative distribution  
21 function. Both results are stated in the small-time representation of the first-  
22 passage time series (Navarro and Fuss, 2009; Gondan et al., 2014), that is,  
23 as a sum over image charges  $r_j$  with Gaussian factors  $\exp(-r_j^2/2t)$ .

24 The starting-point and non-decision-time integrals have remained numer-  
25 ical. Tuerlinckx (2004, pp. 706–708) worked in the large-time representation  
26 of Ratcliff (1978), in which the starting point enters through  $\sin(k\pi w)$ , inte-  
27 grated the starting point analytically at fixed drift, and concluded that the  
28 drift integral then has to be approximated by quadrature. Every current  
29 implementation of the seven-parameter model integrates the starting point  
30 numerically: DMAT (Vandekerckhove and Tuerlinckx, 2008), fast-dm and  
31 rtdists (Voss and Voss, 2007; Singmann et al., 2022), HDDM (Wiecki et al.,  
32 2013), WienR (Hartmann and Klauer, 2021), and the Stan implementation  
33 of Henrich et al. (2024). Blurton et al. (2017) noted that adding starting-  
34 point and non-decision-time variability to their result “requires numerical  
35 evaluation of two integrals” (p. 11).

36 In this note we show that the starting-point integral of the density (1)  
 37 has a closed form. The reason it was missed is representational. In the  
 38 large-time series, after the drift has been integrated, the starting point ap-  
 39 pears in a Gaussian factor with a *positive* coefficient of  $w^2$  and the integral  
 40 against  $\sin(k\pi w)$  lands on the imaginary error function; this is the obstruc-  
 41 tion Tuerlinckx encountered. In the small-time series the factor  $\exp(-r_j^2/2t)$   
 42 contributes a negative coefficient of  $w^2$  that always dominates the positive  
 43 one produced by the drift integral, so the integrand is a linear function times  
 44 a decaying Gaussian, and the integral is a Gaussian moment with a closed  
 45 form. Section 2 gives the result and its derivation, Section 3 the large-time  
 46 counterpart that is needed for numerical stability at large  $t$ , Section 4 the  
 47 truncation rule, Section 5 the partial derivatives, Section 6 the resulting  
 48 seven-parameter likelihood, and Section 7 the numerical verification and a  
 49 comparison of computational cost with existing software.

## 50 2. The density with variable drift and variable starting point

51 Let the starting point be uniformly distributed on  $[w_1, w_2]$ ,  $0 \leq w_1 <$   
 52  $w_2 \leq 1$ . The lower-barrier first-passage time density of the resulting process  
 53 is

$$g(t \mid \nu, \eta^2, a, w_1, w_2) := \frac{1}{w_2 - w_1} \int_{w_1}^{w_2} g(t \mid \nu, \eta^2, a, w) dw. \quad (2)$$

54 Because  $r_j \geq ja$  for  $w \in [0, 1]$ , the  $j$ -th term of (1) is bounded in absolute  
 55 value by  $(j+1)a \exp(-j^2 a^2/2t)$  times a constant independent of  $w$  and  $j$   
 56 (Appendix B), so the series converges absolutely and uniformly on  $[w_1, w_2]$   
 57 and summation and integration may be exchanged. It therefore suffices to  
 58 integrate one term.

### 59 2.1. The $j$ -th term is a Gaussian moment

60 Write  $r_j = p_j + q_j w$  with  $p_j = ja$ ,  $q_j = a$  for even  $j$ , and  $p_j = (j+1)a$ ,  
 61  $q_j = -a$  for odd  $j$ . Collecting the  $w$ -dependence of the  $j$ -th term of (1), that  
 62 is, of  $\exp[(-2\nu aw + \eta^2 a^2 w^2)/2S] r_j \exp(-r_j^2/2t)$ , gives a quadratic exponent  
 63 in  $w$ ,

$$(p_j + q_j w) \exp\left(-\frac{1}{2}Aw^2 + B_j w + C_j\right), \quad (3)$$

64 with

$$A = \frac{a^2}{tS}, \quad B_j = -\frac{\nu a}{S} \mp \frac{k_j a^2}{t}, \quad C_j = -\frac{\nu^2 t}{2S} - \frac{k_j^2 a^2}{2t}, \quad (4)$$

where  $k_j = j$  and the upper sign apply for even  $j$ , and  $k_j = j + 1$  and the lower sign apply for odd  $j$ . The coefficient of  $w^2$  deserves attention. The drift integral contributes  $+\eta^2 a^2/2S$ , a growing Gaussian in  $w$ , whereas the image-charge factor contributes  $-a^2/2t$ , and

$$\frac{a^2}{2t} - \frac{\eta^2 a^2}{2S} = \frac{a^2(S - \eta^2 t)}{2tS} = \frac{a^2}{2tS} > 0 \quad (5)$$

for every  $t > 0$ , every  $\eta$ , and every  $j$ . The combined exponent is therefore a decaying Gaussian with  $A > 0$ . This is the whole content of the result: in the small-time representation the starting-point integrand is a linear function times a proper Gaussian, and such integrals have closed forms in terms of exp and  $\Phi$ .

## 2.2. The result

For  $A > 0$  and  $m := B/A$ ,

$$\begin{aligned} & \int_{w_1}^{w_2} (p + qw) e^{-\frac{1}{2}Aw^2 + Bw + C} dw \\ &= (p + qm) \sqrt{\frac{2\pi}{A}} e^{C+B^2/2A} \left[ \Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m)) \right] \\ & - \frac{q}{A} \left[ e^{C+Bw_2 - \frac{1}{2}Aw_2^2} - e^{C+Bw_1 - \frac{1}{2}Aw_1^2} \right], \end{aligned} \quad (6)$$

where  $\Phi$  denotes the standard normal distribution function (Appendix A). Applying (6) term by term with the coefficients (4) gives the lower-barrier first-passage time density with normally distributed drift and uniformly distributed starting point,

$$\begin{aligned} g(t \mid \nu, \eta^2, a, w_1, w_2) &= \frac{1}{(w_2 - w_1) \sqrt{2\pi t^3 S}} \sum_{j=0}^{\infty} (-1)^j I_j, \\ I_j &= \int_{w_1}^{w_2} (p_j + q_j w) e^{-\frac{1}{2}Aw^2 + B_j w + C_j} dw, \end{aligned} \quad (7)$$

with  $I_j$  given in closed form by (6). The density at the upper barrier is  $g(t \mid -\nu, \eta^2, a, 1 - w_2, 1 - w_1)$ . For  $w_2 - w_1 \rightarrow 0$  the result reduces to (1), and for  $\eta = 0$  it reduces to the starting-point integral of the constant-drift density, so it can be used in a fitting routine regardless of whether either form of variability is present in the data.

Two remarks on evaluation. First, for large  $j$  the centre  $m = B_j/A$  of the Gaussian in (6) lies far outside  $[w_1, w_2]$ , so that  $e^{C+B^2/2A}$  is very large and the bracketed difference of  $\Phi$  values is very small. The product must be formed in logarithmic space, using  $\log \Phi$  and a tail-aware difference; with that precaution the expression is stable for all parameter values we have tested. Second, the identity  $\sqrt{2\pi} e^{C+B^2/2A} \phi(\sqrt{A}(w_i - m)) = e^{C+Bw_i - \frac{1}{2}Aw_i^2}$ , where  $\phi$  is the standard normal density, shows that the two brackets in (6) share their exponential factors; this is used in the derivatives of Section 5.

### 3. Large-time representation

Like every small-time series, (7) loses precision through cancellation when  $t/a^2$  is large; in double precision the loss becomes visible for  $t/a^2 \gtrsim 2$ . A large-time counterpart is obtained from the large-time series of Ratcliff (1978). Integrating  $\exp(-vaw - v^2t/2)$  against the  $N(\nu, \eta^2)$  drift density produces exactly the prefactor of (1), so the large-time density with normally distributed drift is

$$g(t \mid \nu, \eta^2, a, w) = \frac{\pi}{a^2 \sqrt{S}} \exp \left[ \frac{-\nu^2 t - 2\nu a w + \eta^2 (aw)^2}{2S} \right] \times \sum_{k=1}^{\infty} k e^{-k^2 \pi^2 t / 2a^2} \sin(k\pi w). \quad (8)$$

Here there is no image-charge factor to compensate the growing Gaussian, and the starting-point integral of the  $k$ -th term is  $\int_{w_1}^{w_2} e^{\kappa w^2 + \lambda w} \sin(k\pi w) dw$  with  $\kappa = \eta^2 a^2 / 2S > 0$  and  $\lambda = -\nu a / S$ . Written as the imaginary part of  $\int e^{\kappa w^2 + \mu_k w} dw$  with  $\mu_k = \lambda + ik\pi$ , the antiderivative is an imaginary error function of complex argument, which overflows. It is, however, bounded when expressed through the Faddeeva function  $w(z) = e^{-z^2} \operatorname{erfc}(-iz)$  (Abramowitz and Stegun, 1964, 7.1.3):

$$\int_{w_1}^{w_2} e^{\kappa w^2 + \mu_k w} dw = \frac{-i\sqrt{\pi}}{2\sqrt{\kappa}} \left[ e^{\kappa w^2 + \mu_k w} w(z_k(w)) \right]_{w_1}^{w_2}, \quad z_k(w) = \sqrt{\kappa} w + \frac{\mu_k}{2\sqrt{\kappa}}. \quad (9)$$

Since  $\operatorname{Im} z_k = k\pi / 2\sqrt{\kappa} > 0$ ,  $w(z_k)$  lies in the upper half plane where it is bounded by one, and the product with  $e^{\kappa w^2 + \mu_k w}$  is of the size of the integrand. For  $\eta = 0$  the bracket reduces to  $(e^{\mu_k w_2} - e^{\mu_k w_1}) / \mu_k$ . The large-time density

110 with uniformly distributed starting point is then

$$g(t \mid \nu, \eta^2, a, w_1, w_2) = \frac{\pi e^{-\nu^2 t/2S}}{a^2 \sqrt{S} (w_2 - w_1)} \sum_{k=1}^{\infty} k e^{-k^2 \pi^2 t/2a^2} \operatorname{Im} \int_{w_1}^{w_2} e^{\kappa w^2 + \mu_k w} dw. \quad (10)$$

111 Fast and accurate algorithms for  $w(z)$  are standard (Zaghloul and Ali, 2011)  
 112 and available in scientific libraries. In our implementation the two represen-  
 113 tations are switched at  $t/a^2 = 1$ ; they agree to  $1.5 \times 10^{-11}$  relative across  
 114 the switch. We stress that (9) must be used in the Faddeeva form: written  
 115 with the imaginary error function it loses of the order of  $\pi^2 k^2/4\kappa$  nats to  
 116 cancellation and returns meaningless values when the series is truncated at  
 117 a fixed large  $K$ .

#### 118 4. Convergence

119 The  $j$ -th term of (7) is bounded in absolute value by  $c(j +$   
 120  $1)a \exp(-j^2 a^2/2t)$ , and the  $k$ -th term of the large-time series (10) by  
 121  $c' k \exp(-k^2 \pi^2 t/2a^2)$ , where the constants  $c$  and  $c'$  do not depend on the  
 122 index (Appendix B). Both bounds decrease super-exponentially, so the trun-  
 123 cation error is of the order of the first neglected term, and for a tolerance  $\varepsilon$   
 124 the series may be cut at

$$J = \left\lceil \frac{\sqrt{2t \log(1/\varepsilon)}}{a} \right\rceil + 1, \quad K = \left\lceil a \sqrt{\frac{2 \log(1/\varepsilon)}{\pi^2 t}} \right\rceil + 1. \quad (11)$$

125 With  $\varepsilon = 10^{-12}$  the small-time series needs  $J \leq 9$  terms for  $t/a^2 \leq 1$ , and  
 126 the large-time series needs  $K \leq 4$  terms for  $t/a^2 \geq 1$ . Evaluating (7) with  
 127  $\varepsilon = 10^{-12}$  and with  $\varepsilon = 10^{-40}$  on the 400 parameter sets of Section 7 gives  
 128 results that differ by at most  $2.2 \times 10^{-16}$ .

#### 129 5. Partial derivatives

130 Gradient information is required by Hamiltonian Monte Carlo and by  
 131 gradient-based optimisers, and Hartmann and Klauer (2021) derived it for  
 132 the constant-parameter model. Every term of (7) and (10) is a composition  
 133 of  $\exp$ ,  $\Phi$ , and  $w$ , so the partial derivatives with respect to  $\nu$ ,  $\eta$ ,  $a$ ,  $w_1$ , and  $w_2$   
 134 are again closed form. For the small-time representation it is convenient to  
 135 differentiate  $I_j$  with respect to its seven coefficients and to apply the chain

136 rule. Writing  $G := \sqrt{2\pi/A} e^{C+B^2/2A} [\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m))]$   
 137 and  $E_i := e^{C+Bw_i-\frac{1}{2}Aw_i^2}$ , so that  $I = (p + qm)G - (q/A)(E_2 - E_1)$ , the total  
 138 differential of  $I$  under  $(dA, dB, dC, dp, dq, dw_1, dw_2)$  is

$$\begin{aligned} dI &= (dp + m dq + q dm) G + (p + qm) dG \\ &\quad - \left( \frac{dq}{A} - \frac{q dA}{A^2} \right) (E_2 - E_1) - \frac{q}{A} (dE_2 - dE_1), \\ dm &= \frac{dB - m dA}{A}, \quad dK = dC + m dB - \frac{1}{2}m^2 dA, \\ dG &= G \left( dK - \frac{dA}{2A} \right) + E_2 \left[ (dw_2 - dm) + \frac{(w_2 - m) dA}{2A} \right] \\ &\quad - E_1 \left[ (dw_1 - dm) + \frac{(w_1 - m) dA}{2A} \right], \\ dE_i &= E_i \left[ dC + w_i dB - \frac{1}{2}w_i^2 dA + (B - Aw_i) dw_i \right], \end{aligned} \quad (12)$$

139 in which the identity noted at the end of Section 2 has been used to express  
 140  $dG$  through  $E_1$  and  $E_2$  rather than through  $\phi$ . The coefficient differentials  
 141 follow from (4); for instance  $\partial B_j / \partial \nu = -a/S$  and  $\partial C_j / \partial \nu = -\nu t/S$ , and  
 142  $A, p_j, q_j$  do not depend on  $\nu$ . For the large-time representation only two  
 143 identities are needed, both immediate from  $\frac{d}{dw} e^{\kappa w^2 + \mu w} = (2\kappa w + \mu) e^{\kappa w^2 + \mu w}$ :  
 144 with  $\mathcal{I} := \int_{w_1}^{w_2} e^{\kappa w^2 + \mu w} dw$  and  $\mathcal{E}_i := e^{\kappa w_i^2 + \mu w_i}$ ,

$$\frac{\partial \mathcal{I}}{\partial \mu} = \frac{(\mathcal{E}_2 - \mathcal{E}_1) - \mu \mathcal{I}}{2\kappa}, \quad \frac{\partial \mathcal{I}}{\partial \kappa} = \frac{(w_2 \mathcal{E}_2 - w_1 \mathcal{E}_1) - \mathcal{I} - \mu \partial \mathcal{I} / \partial \mu}{2\kappa}. \quad (13)$$

145 The derivatives with respect to  $w_1$  and  $w_2$  are, by the fundamental theorem  
 146 of calculus, the integrand evaluated at the endpoints, corrected for the fac-  
 147 tor  $1/(w_2 - w_1)$ . All of (12) and (13), and every chain-rule coefficient, are  
 148 verified as exact symbolic identities in the supplementary code; the resulting  
 149 gradients agree with those of Hartmann and Klauer (2021) as implemented  
 150 in WienR to  $3 \times 10^{-13}$  absolute on the parameter grid of Section 7.

## 151 6. The seven-parameter likelihood

152 With the non-decision time uniformly distributed on  $[t_0, t_0 + s_t]$ , the den-  
 153 sity of an observed response time is

$$f(t) = \frac{1}{s_t} \int_{t-t_0-s_t}^{t-t_0} g(u \mid \nu, \eta^2, a, w_1, w_2) du, \quad (14)$$

Table 1: Maximum relative error of Gauss–Legendre quadrature for (14) against adaptive quadrature, 200 random parameter sets, densities above  $10^{-3}$ .

| Nodes | Plain                | With cubic map       |
|-------|----------------------|----------------------|
| 8     | $4.1 \times 10^{-3}$ | $4.3 \times 10^{-3}$ |
| 16    | $7.3 \times 10^{-5}$ | $3.5 \times 10^{-6}$ |
| 24    | $4.1 \times 10^{-6}$ | $2.7 \times 10^{-8}$ |
| 32    | $1.1 \times 10^{-8}$ | $1.5 \times 10^{-9}$ |

where the lower limit is replaced by 0 if negative. This integral does not have a closed form: the non-decision time shifts the time argument, which enters every term of (7) through  $\sqrt{t}$  and  $1/t$ , and we have not found a representation in which the shift integrates. What (7) achieves is to reduce the seven-parameter likelihood from a two-dimensional cubature over  $(w, u)$  to a one-dimensional quadrature over  $u$  of a smooth closed-form integrand on a short interval. Because  $g(u)$  rises steeply from zero at  $u = 0$  and the window may include that leading edge, a plain Gauss–Legendre rule converges slowly; the substitution  $u = u_{lo} + (u_{hi} - u_{lo})s^3$ , which clusters nodes at the lower limit, fixes this. Table 1 gives the maximum relative error of the rule against adaptive quadrature over the 200 seven-parameter sets described below. A 32-node rule with the cubic map reaches  $1.5 \times 10^{-9}$  at a fixed cost of 32 evaluations of (7). The gradient of (14) with respect to the five parameters of  $g$  is the same rule applied to the gradient of  $g$ , and  $\partial f / \partial t_0 = [g(u_{lo}) - g(u_{hi})] / s_t$  and  $\partial f / \partial s_t = [g(u_{lo}) - f] / s_t$  follow from the endpoints.

## 7. Verification and efficiency

### 7.1. Verification

We checked the result in five independent ways. (i) Every algebraic step, in particular the antiderivative (6), the inequality (5), the coefficient table (4), the drift integral leading to (1), and all derivative formulae of Section 5, is verified as an exact symbolic identity in a computer algebra system, that is, as  $\text{simplify}(\text{d}F/\text{d}x - f) = 0$  rather than by numerical agreement. (ii) In 40- to 60-digit arithmetic, (7) agrees with a brute-force double integral of the constant-parameter density over  $(v, w)$  to  $8.7 \times 10^{-32}$ , and (10) agrees with quadrature of (1) to between  $10^{-35}$  and  $10^{-48}$  at  $t/a^2 = 0.9, 1.5$ , and  $4.7$ .

180 (iii) On 400 random parameter sets with  $a \in \{0.6, 1, 1.5, 2.5\}$ ,  $\nu \in [-3, 4]$ ,  
 181  $\eta \in \{0.5, 1, 2\}$ ,  $w_2 - w_1 \in \{0.05, 0.2, 0.4\}$ , and  $t \in [0.05, 3]$ , spanning densi-  
 182 ties from  $5 \times 10^{-20}$  to 1.6, the switched representation agrees with WienR  
 183 (Hartmann and Klauer, 2021) called at its tightest tolerance ( $10^{-12}$ ) to a  
 184 maximum absolute difference of  $1.2 \times 10^{-14}$ ; rtdists (Singmann et al., 2022),  
 185 whose diffusion code is that of fast-dm, converges onto the same values as  
 186 its precision setting is raised ( $7.6 \times 10^{-4}$ ,  $1.1 \times 10^{-5}$ ,  $2.5 \times 10^{-8}$  at precision  
 187 3, 5, 8), as does the HDDM likelihood (Wiecki et al., 2013) as its integra-  
 188 tion tolerance is tightened. On 200 seven-parameter sets, (14) agrees with  
 189 WienR to  $2.5 \times 10^{-9}$ , which is WienR’s own reported cubature error. (iv) An  
 190 Euler–Maruyama simulation of the process with 400 000 trials, step  $5 \times 10^{-5}$ ,  
 191 and the Broadie–Glasserman–Kou barrier correction gives a lower-barrier re-  
 192 sponse time histogram in 30 bins of 100 ms whose  $\chi^2$  against the binned  
 193 density (7) is 21.6, with all standardised residuals below 1.9. (v) The deriva-  
 194 tion of (7), including the antiderivative, the coefficient table, the exchange of  
 195 summation and integration, and the Faddeeva integral (9), is formalised and  
 196 machine-checked in Lean 4 with Mathlib (de Moura and Ullrich, 2021; The  
 197 mathlib Community, 2020); the development contains no unproved state-  
 198 ment and relies on no axioms beyond Lean’s standard ones. The starting  
 199 representations (1) and (8) are taken as definitions there, as they are taken  
 200 from the literature here.

## 201 7.2. Efficiency

202 Table 2 compares the cost of (7) and (14), compiled from a direct tran-  
 203 scription of the formulae, with the three packages in common use: WienR,  
 204 which evaluates the drift-integrated series by adaptive cubature over  $w$ ; rt-  
 205 dists, whose diffusion code is that of fast-dm and integrates over  $w$  and the  
 206 non-decision window by adaptive Simpson’s rule; and the HDDM likelihood,  
 207 which does the same with the Navarro–Fuss series. All timings are single-  
 208 threaded on one machine,<sup>2</sup> for 10 000 response times at one parameter set,  
 209 which is the situation in a likelihood evaluation.

210 The error columns need a reference scale. Blurton et al. (2017) recom-  
 211 mend a truncation tolerance of  $1.5 \times 10^{-8}$ , the square root of single-precision  
 212 machine epsilon, and Gondan et al. (2014) illustrate their term counts at

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<sup>2</sup>Apple M1 Pro; R 4.5.3 with WienR 0.3-15 and rtdists 0.11-6; Python 3.11 with NumPy 1.26.4, SciPy 1.13.1, Cython 3.1.4, and hddm-wfpt 0.1.4, the HDDM likelihood distributed as a standalone package.

Table 2: Cost per evaluation in microseconds, single thread, 10 000 response times at  $a = 1.2$ ,  $\nu = 1$ ,  $w = 0.5$ ,  $\eta = 1$ ,  $w_2 - w_1 = 0.2$ ,  $s_t = 0.2$ . Each package at the settings its fitting routine uses: WienR at tolerance  $10^{-12}$ , rtdists at precision 3, HDDM at series tolerance  $10^{-4}$  with Simpson tolerance  $10^{-8}$  and recursion depth 10. Error is the maximum relative deviation from (7) or (14) over the parameter sets of Section 7 (densities above  $10^{-3}$ ); for the closed form the seven-parameter entry is the quadrature error of (14) against adaptive integration. <sup>†</sup>Uncompiled vectorised implementation.

|                                        | Six parameters        |         | Seven parameters (+ $s_t$ ) |         |
|----------------------------------------|-----------------------|---------|-----------------------------|---------|
|                                        | Error                 | $\mu s$ | Error                       | $\mu s$ |
| Closed form                            | –                     | 0.33    | $1.5 \times 10^{-9}$        | 9.2     |
| WienR                                  | $2.5 \times 10^{-12}$ | 2.1     | $1.5 \times 10^{-9}$        | 1040    |
| rtdists                                | $4.6 \times 10^{-3}$  | 0.43    | $9.2 \times 10^{-3}$        | 18      |
| HDDM                                   | $3.9 \times 10^{-6}$  | 0.60    | $1.5 \times 10^{-7}$        | 46      |
| Closed form + 5 gradients <sup>†</sup> | –                     | 1.7     |                             |         |
| WienR, 5 gradients                     | $10^{-13}$            | 15.7    |                             |         |

213  $10^{-3}$ . Tuerlinckx (2004, Table 1) reports differences of  $2 \times 10^{-7}$  to  $3 \times 10^{-4}$   
214 between his quadrature-based algorithm and the classical one. Among cur-  
215 rent packages, rtdists (fast-dm) defaults to a precision of 3, documented as  
216 roughly the number of accurate decimals; the HDDM likelihood truncates  
217 its series at  $10^{-4}$  and integrates over the starting point and the non-decision  
218 window by adaptive Simpson’s rule with tolerance  $10^{-8}$  and recursion depth  
219 10; the Stan implementation of Henrich et al. (2024) fixes the density tol-  
220 erance at  $10^{-6}$  and defaults to  $10^{-4}$  for the derivatives; WienR defaults to  
221  $10^{-12}$ . The closed form is exact to double precision, and its seven-parameter  
222 quadrature at  $1.5 \times 10^{-9}$  is below every one of these tolerances.

223 Two features of Table 2 are worth stating. First, the closed form is the  
224 cheapest of the four at every entry and the only one that is exact for six  
225 parameters. rtdists costs 1.3 times as much for six parameters and twice as  
226 much for seven, and deviates from the density by 0.5% and 0.9%. HDDM  
227 costs 1.8 and 5 times as much and agrees to  $4 \times 10^{-6}$  and  $1.5 \times 10^{-7}$ ; its  
228 series tolerance of  $10^{-4}$  leaves a floor near  $10^{-7}$  that tightening its integration  
229 settings does not remove. WienR reaches  $10^{-12}$  at 6 and 110 times the cost.  
230 Because the cost of the closed form is fixed whereas that of the integrators  
231 is set by the accuracy asked of them, these ratios grow with the tolerance.  
232 Second, the gradient comes almost free: all five partial derivatives of (7) share  
233 the exponential factors of the density itself, and an uncompiled vectorised

234 implementation returns the density together with all five derivatives in  $1.7 \mu\text{s}$ ,  
 235 whereas WienR requires one further cubature per derivative. For reference,  
 236 WienR’s own closed form for drift variability alone (Blurton et al., 2017)  
 237 evaluates in  $0.26 \mu\text{s}$  on the same machine, so the closed form with starting-  
 238 point variability costs 1.3 times the drift-only density.

## 239 8. Discussion

240 We have shown that the first-passage time density of the diffusion model  
 241 with normally distributed drift and uniformly distributed starting point has a  
 242 closed form, and that the reason it was not found earlier is the choice of series  
 243 representation: the small-time series carries a Gaussian factor in the starting  
 244 point that the large-time series lacks, and that factor is exactly what gives  
 245 the integral a closed form. The result completes the analytic treatment of  
 246 the drift and starting-point integrals of the seven-parameter model and leaves  
 247 only the non-decision-time integral, which is a one-dimensional quadrature  
 248 of a smooth function on a short interval and for which we have given a fixed  
 249 32-node rule accurate to  $10^{-9}$ .

250 For practice the consequences are threefold. The density is exact up to  
 251 a truncation error that is controlled by (11); it is evaluated at a fixed cost  
 252 that is below that of each existing package at the settings the package fits  
 253 with; and its partial derivatives are closed form and nearly free, which is  
 254 what hierarchical Bayesian and Hamiltonian Monte Carlo implementations  
 255 (Wiecki et al., 2013; Henrich et al., 2024) need. The result is directly usable  
 256 in any of the existing packages, which already contain the drift-integrated  
 257 density (1) as a building block and need only replace the quadrature over  $w$   
 258 by (6).

259 The corresponding cumulative distribution function with both forms of  
 260 variability does not follow by the same route. Starting from the drift-  
 261 integrated distribution function of Blurton et al. (2017), half of the series  
 262 terms have a coefficient of  $w^2$  that is exactly zero after the drift integral and  
 263 close through the integral  $\int e^{cw} \Phi(\alpha w + \beta) dw$ , which has a closed form; the  
 264 other half carry the coefficient  $+2a^2\eta^2$ , a growing Gaussian times  $\Phi$ , and  
 265 reduce to an integral of Owen- $T$  type that has no closed form in terms of  
 266  $\exp$ ,  $\Phi$ , or Owen’s  $T$ . For maximum likelihood and Bayesian estimation the  
 267 density suffices, and we leave the distribution function open.

## 268 Declaration of competing interest

269 The author declares no competing interests.

## 270 Data availability

271 All code, reference values, comparison scripts, and the Lean 4 formalisa-  
272 tion are available in the supplementary material and at [repository URL to  
273 be added on acceptance].

## 274 Funding

## 275 Declaration of generative AI in the writing process

## 276 Appendix A. The antiderivative

277 Let  $A > 0$ ,  $m = B/A$ , and

$$F(w) := (p + qm) \sqrt{\frac{2\pi}{A}} e^{C+B^2/2A} \Phi(\sqrt{A}(w - m)) - \frac{q}{A} e^{C+Bw-\frac{1}{2}Aw^2}.$$

278 Using  $\Phi' = \phi$  and  $-\frac{1}{2}A(w - m)^2 + C + B^2/2A = C + Bw - \frac{1}{2}Aw^2$ ,

$$\begin{aligned} F'(w) &= (p + qm) \sqrt{2\pi} e^{C+B^2/2A} \phi(\sqrt{A}(w - m)) - \frac{q}{A} (B - Aw) e^{C+Bw-\frac{1}{2}Aw^2} \\ &= [(p + qm) - q(m - w)] e^{C+Bw-\frac{1}{2}Aw^2} = (p + qw) e^{C+Bw-\frac{1}{2}Aw^2}, \end{aligned}$$

279 which is the integrand of (6). The definite integral (6) is  $F(w_2) - F(w_1)$ .  
280 The same computation is carried out symbolically in the supplementary code  
281 and formally in the Lean development.

## 282 Appendix B. Term bounds

283 For  $w \in [0, 1]$ ,  $ja \leq r_j \leq (j + 1)a$ , and the prefactor of (1) is bounded  
284 by  $\exp[(2|\nu|a + \eta^2 a^2)/2S]$ . Hence the  $j$ -th term of the series in (1), and  
285 therefore  $|I_j|/(w_2 - w_1)$  in (7), is at most  $(j + 1)a \exp(-j^2 a^2/2t)$  times a  
286 constant independent of  $j$  and  $w$ . Since  $\sum_j (j + 1) \exp(-j^2 c)$  converges for  
287 every  $c > 0$ , the series converges absolutely and uniformly in  $w$ , which justifies  
288 the exchange of summation and integration and gives the truncation rule for  
289  $J$  in (11). For the large-time series (10),  $|w(z)| \leq 1$  in the upper half plane  
290 and  $|e^{\kappa w^2 + \mu_k w}| \leq e^{\kappa + |\lambda|}$  on  $[0, 1]$ , so the  $k$ -th term is at most a constant times  
291  $k \exp(-k^2 \pi^2 t/2a^2)$ , which gives the rule for  $K$ .

## 292 **Appendix C. Supplementary material**

293 The supplementary material contains (i) a vectorised implementation  
294 of (7), (10), (14), and all partial derivatives in Python, together with a  
295 compiled kernel used for Table 2; (ii) the symbolic verification script; (iii)  
296 the scripts and reference values for the comparisons with WienR, rtdists, and  
297 HDDM, and the simulation; and (iv) the Lean 4 formalisation.

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